Experiment No. 21 – Opening Morphological Operation

1. Aim

To implement the Opening technique as a Morphological Operation using OpenCV to remove small noise from the foreground regions.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Image loading failed")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Convert to binary image
_, binary = cv2.threshold(
    gray, 127, 255, cv2.THRESH_BINARY
)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply opening operation
opening = cv2.morphologyEx(
    binary, cv2.MORPH_OPEN, kernel
)

# Display output
plt.figure(figsize=(12, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(binary, cmap="gray")
plt.title("Binary Image")
plt.axis("off")
```

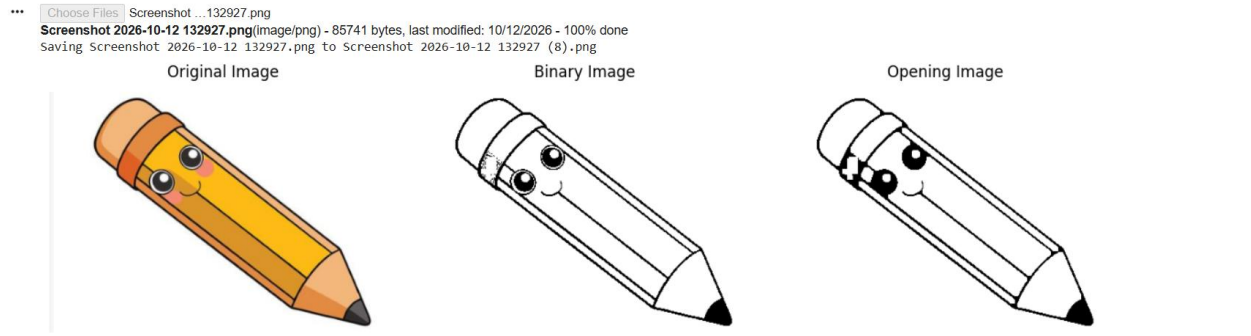
```
plt.subplot(1, 3, 3)
plt.imshow(opening, cmap="gray")
plt.title("Opening Image")
plt.axis("off")


plt.tight_layout()
plt.show()
```

3. Expected Output

The output shows that small white noise particles are removed while larger foreground objects are preserved.

4. Output



The opening operation is successfully applied to remove small unwanted foreground objects.

5. Result

Thus, the Opening Morphological Operation was successfully implemented using Python and OpenCV.